

Technology Stack

Date	27 JUNE 2026
Team ID	xxxxxx
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	Streamlit, HTML, CSS	Streamlit provides an interactive and user-friendly interface for users to manage loans, analyze financial health, and view AI-generated recommendations. HTML and CSS improve UI design and responsiveness.

2	Backend / Server-Side	Python, FastAPI	FastAPI provides high-performance REST APIs for user authentication, loan management, financial analysis, and AI integration with fast request processing.
3	Database / Data Storage	SQLite, SQLAlchemy	SQLite securely stores user accounts, loan information, repayment history, and settlement data. SQLAlchemy simplifies database operations using ORM.
4	Cloud / Hosting / Deployment	Uvicorn, Local Server (Deployable on AWS/GCP/Azure)	Uvicorn runs the FastAPI application efficiently. The platform can later be deployed on cloud services for scalability and availability.
5	Version Control & CI/CD	Git, GitHub	Git and GitHub are used for source code management, collaboration, version control, project backup, and continuous development.

6	Third-Party APIs / Other Tools	AI Model API (Gemini/OpenAI), Credit Bureau APIs, Banking APIs	AI APIs generate financial analysis and debt settlement recommendations. Credit Bureau and Banking APIs retrieve financial data for accurate analysis and personalized recovery plans.
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